

Question Dataset

SOLAR SYSTEM - Factual (Q1–Q5)

Q1. What are the two main chemical elements that make up Jupiter and Saturn?

Reference answer: Hydrogen and helium

Q2. What is the surface temperature range on Mercury?

Reference answer: 280 to 430 degrees Celsius on its sunlit side

Q3. Which planet has a thick carbon dioxide atmosphere that keeps its surface temperature at around 700 K?

Reference answer: Venus

Q4. How old is the solar system?

Reference answer: Approximately 4.5 billion years old

Q5. What is the name of Jupiter's moon that may have an ocean under its frozen ice?

Reference answer: Europa

SOLAR SYSTEM - Conceptual (Q6–Q10)

Q6. Why are Jupiter and Saturn sometimes misleadingly called gas planets?

Reference answer: Because the gas in their interiors is compressed by gravity until the hydrogen becomes a liquid, meaning the bulk of both planets consists of compressed liquefied hydrogen rather than gas

Q7. Why does Venus have higher surface temperatures than Mercury despite being farther from the Sun?

Reference answer: Because Venus has a thick atmosphere of carbon dioxide that acts as insulation, trapping heat at the surface through the greenhouse effect, whereas Mercury has no atmosphere to retain heat

Q8. Why is Earth the only planet in the solar system that can support life as we know it?

Reference answer: Because Earth is the only planet where surface temperatures generally lie between the freezing and boiling points of water, making liquid water possible

Q9. Why do asteroids and comets help scientists understand the early solar system?

Reference answer: Because most asteroids and comets retain their original composition and were never heated to the melting point, making them relatively unmodified material from the time of the solar system's formation

Q10. Why does a planet's distance from the Sun affect its surface temperature?

Reference answer: Because planets are heated by radiant energy from the Sun, which gets weaker with the square of the distance, meaning the farther a planet is from the Sun the less energy it receives and the cooler its surface

SOLAR SYSTEM - Document-dependent (Q11–Q15)

Q11. According to the document, what is the term used for the category of objects invented to include Pluto and similar icy objects beyond Neptune?

Reference answer: Dwarf planet

Q12. According to the document, what event in 1994 provided dramatic evidence that planetary bombardment continues today?

Reference answer: The collision of more than 20 large pieces of Comet Shoemaker-Levy 9 with Jupiter

Q13. According to the document, what spacecraft missions are mentioned as having landed on or orbited planets and moons in the solar system?

Reference answer: Voyager, Pioneer, Curiosity, and Pathfinder

Q14. According to the document, what is the surface temperature on Pluto, and how does it compare to Mercury?

Reference answer: Pluto's temperature is about -220 degrees Celsius, which is colder than liquid air, and is less than that of Mercury by a factor of 10

Q15. According to the document, why would the oceans of Earth be permanently frozen without its atmosphere?

Reference answer: Because without atmospheric insulation from the greenhouse effect, Earth would not retain enough heat to keep its oceans liquid

ROMAN EMPIRE - Factual (Q16–Q20)

Q16. What was the Latin term for the government's official distribution of grain to the Roman populace?

Reference answer: The annona

Q17. What were large agricultural estates in the Roman countryside known as?

Reference answer: Latifundia

Q18. What was the dining room in a Roman home called?

Reference answer: The triclinium

Q19. What was the formal process by which deceased emperors were made gods called?

Reference answer: Apotheosis

Q20. What were the large storage vessels used to ship wine and olive oil across the Roman Empire called?

Reference answer: Amphorae

ROMAN EMPIRE - Conceptual (Q21–Q25)

Q21. Why was the layout of a wealthy Roman home designed to impress visitors?

Reference answer: Because the floor plan emphasised the power relationship between a patron and his clients, as well as the authority and prestige of the paterfamilias, with fountains, gardens, and lavish features positioned so guests could see them upon arrival

Q22. Why was the annona both an economic and a political tool for Roman emperors?

Reference answer: Because while it kept the populace fed, it also allowed emperors to associate themselves with generosity and build public support, meaning the distribution of grain was tied to the personality and image of the emperor

Q23. Why did the Roman Empire rely so heavily on Egypt for its grain supply?

Reference answer: Because Egypt functioned as the empire's breadbasket, producing grain on a large scale, which freed Italian farmers to focus on higher-priced agricultural products like wine and olive oil

Q24. Why was the imperial cult significant in Roman religious and political life?

Reference answer: Because it connected religious practice with political loyalty by deifying deceased emperors, making worship of the emperor part of public life and allowing priesthoods attached to the cult to advance men's public careers

Q25. Why were the navicularii compelled by the Roman government to join collegia?

Reference answer: So they could be easily supervised, and in exchange for signing contracts to supply grain they received benefits including exemption from other public service

ROMAN EMPIRE - Document-dependent (Q26–Q30)

Q26. According to the document, from what regions did the Roman Empire source silk and furs through the monsoon-driven Indian Ocean trade network?

Reference answer: Silk came from China and India, and furs came from the Baltic region

Q27. According to the document, which emperor ruled from 138 to 161 CE and used the annona to create a positive public image?

Reference answer: Hadrian

Q28. According to the document, what was the role of the praefectus annonae?

Reference answer: The prefect who oversaw the grain distribution process, governed the ports to which grain was shipped, addressed fraud in the market, and secured the grain supply from Egypt and other regions by signing contracts with suppliers

Q29. According to the document, what happened to glass manufacturing in central Italy in the second century CE?

Reference answer: Manufacturing in Gaul (present-day France) and Germany took over most of its production

Q30. According to the document, what volcanic event preserved the city of Pompeii and made it a key source of information about daily Roman life?

Reference answer: The eruption of Mount Vesuvius in 79 CE